

PRAYCG Formula Module Report v1.1

Full mathematical specification for the Master Comprehensive Analysis Suite v1.5.6 / DGA update

Status: working formula report, public-facing but technically detailed.

Scope: PRAYCG2.0 / Master Comprehensive Analysis Suite v1.5.6, including A-MRED, MRED-Peak/Resolution, NIP/CET/EET, NUPI, TTI, MRED-ITP, OCM/RSM/CVB/Squint, Topo-OSM, visualizer outputs, offline interpreter, and Opto-PING synthetic-gate references.

Central boundary: these formulas define operational indices and analysis gates. They do not prove consciousness, OSM biology, cellular hidden variables, microtubules, biophotons, memory formation, morality, or literal thermodynamic heat/energy flow.

1. Purpose of this report

This report turns the PRAYCG Master Comprehensive Analysis Suite into a formula catalog. It explains each module in enough mathematical detail that a user can understand what the suite is doing before trusting any output table.

The suite currently contains more modules than a confirmatory paper should emphasize. The recommended endpoint hierarchy is:

BOXED PRIMARY PATH:

Timing/QC

+ StimulusFingerprint / CET-R

+ artifact and confound gates

-> A-MRED / MRED-Peak-Resolution

SECONDARY SUMMARY LAYERS:

NIP / BIT / CII / IAQ

TTI

NUPI

DGA / Decoder Gate Availability

Baseline1 vs Baseline2

EXPLORATORY / CONVERGENCE LAYERS:

KHT-topo

NAST

EET

MRED-ITP / ACG / OCU

OCM025 / RSM / CVB / SquintProxy

LSO / Subtitle Override

Opto-PING / Rung 0 synthetic gates

The purpose of this formula report is not to make all modules equal. It is to make every module auditable.

2. Global notation

2.1 Conditions and phases

C = phase-scrambled Control

T = intact Target watched naturally

O = Contextual Override: same intact narrative, analytic task stance

B1 = Baseline 1

W1, W2, W3 = washout windows after Control, Target, Override

B2 = final reflection baseline / Baseline 2

A time sample is indexed by t. A window or anchor is indexed by j. A cue is indexed by i. A frequency band is indexed by b.

2.2 Input streams

x_EEG,ch(t) = raw or cleaned EEG for channel ch

r(t) = R-R intervals or heart-rate stream

resp(t) = respiration stream

u(t) = exogenous stimulus vector

m(t) = LSL / Stasis marker stream

s(t) = self-report / report metadata stream

The exogenous stimulus vector is:

$u(t) = [\text{audio_env}(t), \text{luminance}(t), \text{visual_change}(t), \text{optical_flow}(t), \text{cut_train}(t), \text{cue_train}(t), \text{cue_value}(t), \text{ALS_pulse}(t), \text{anchor_design}(t)]$

Boundary rule:

u(t) belongs to the stimulus / environment side.

It is not biological hidden Y(t).

2.3 Robust z-score

Most module scores use standardized components. The default robust z-score is:

$rz(x_t) = (x_t - \text{median_ref}(x)) / (1.4826 * \text{MAD_ref}(x) + \text{eps})$

$\text{MAD_ref}(x) = \text{median_ref}(|x_t - \text{median_ref}(x)|)$

eps prevents division by zero.

A conventional z-score is:

$$z(x_t) = (x_t - \text{mean_ref}(x)) / (\text{std_ref}(x) + \text{eps})$$

The reference distribution may be:

baseline only

all non-artifact windows

condition-specific windows

random-anchor null windows

pooled Control/Target/Override windows

The reference set must be declared by the module.

2.4 Positive clipping and logistic transform

$$\text{clip_plus}(x) = \max(0, x)$$

$$\text{sigmoid}(x) = 1 / (1 + \exp(-x))$$

Positive clipping is used when a module should count only positive evidence and not allow negative evidence to cancel unrelated terms.

2.5 Pre / hit / post windows

For an anchor at time a_j :

$$W_{\text{pre},j} = [a_j - T_{\text{pre}}, a_j)$$

$$W_{\text{hit},j} = [a_j - T_{\text{left}}, a_j + T_{\text{right}}]$$

$$W_{\text{post},j} = (a_j + \text{lag_start}, a_j + \text{lag_end}]$$

Typical values:

$$T_{\text{pre}} = 10 \text{ to } 30 \text{ s}$$

$$T_{\text{left}}/T_{\text{right}} = 2 \text{ to } 10 \text{ s around anchor}$$

$$\text{lag_start} = 5 \text{ to } 10 \text{ s}$$

$$\text{lag_end} = 20 \text{ to } 30 \text{ s}$$

The exact values should be module-specific and predeclared for strict runs.

3. Data ingestion and timebase reconstruction

3.1 Marker-defined phase slicing

The runner writes marker pairs such as:

TARGET_1_START, TARGET_1_END
CONTROL_1_START, CONTROL_1_END
CONTEXTUAL_OVERRIDE_1_START, CONTEXTUAL_OVERRIDE_1_END
BASELINE_1_START, BASELINE_1_END
BASELINE_2_REFLECTION_START, BASELINE_2_REFLECTION_END

For each phase p:

phase_interval_p = [m_start,p, m_end,p]

A sample belongs to phase p if:

$m_start,p \leq t_sample < m_end,p$

3.2 Continuous-stream reconstruction

If raw LSL timestamps are irregular, the suite can reconstruct time from status markers or sample index:

$t_hat_n = t_anchor + (n - n_anchor) / fs_eff$

$fs_eff = (n_2 - n_1) / (t_2 - t_1)$

where (n_1, t_1) and (n_2, t_2) are sample-count / LSL-time anchor pairs. The reconstructed axis is used only when it improves consistency and is logged as a timing caveat.

3.3 Branch-relative time

For rendering and anchor analysis:

$t_rel = t_abs - t_phase_start$

Event overlays should be repaired when mixed timebases are detected:

if event_time > phase_duration and phase_start_abs is known:

 event_time_rel = event_time - phase_start_abs

else:

 event_time_rel = event_time

4. MediaPrep formulas

MediaPrep prepares the stimulus suite. It does not certify physiology.

4.1 Cue schedule

For fixed-interval number cues:

$$t_i_start = t_start_delay + (i - 1) * T_cue_interval$$

$$t_i_end = t_i_start + T_display$$

where typical PRAYCG settings are:

$$T_cue_interval = 3.0 \text{ s}$$

$$T_display = 0.85 \text{ s}$$

$$cue_position = upper_right$$

The running sum is:

$$S_i = S_{\{i-1\}} + v_i$$

$$S_0 = 0$$

where v_i is the cue value.

4.2 Target/Override identity rule

The Target and Override media are generated from the same cue-embedded stimulus:

$$\text{hash}(\text{Target_video}) = \text{hash}(\text{Override_video})$$

The only intended difference is participant instruction:

Target instruction: watch naturally.

Override instruction: perform analytic cue task.

4.3 Phase-scrambled Control

A simplified visual phase-scramble operation can be written in Fourier form:

$$F_target(k) = A(k) * \exp(i * \phi(k))$$

$$F_control(k) = A(k) * \exp(i * \phi_random(k))$$

$$\text{control_frame} = \text{inverse_Fourier}(F_control)$$

The intent is:

preserve low-level audiovisual structure

remove recognizable narrative meaning

The Control is generated from the cue-embedded Target so the cue rhythm and cue energy remain represented.

4.4 ALS / photodiode stimulus pulse

The ALS timing square is an exogenous display-timing regressor:

$$ALS_pulse(t) = 1 \text{ if } t \text{ in } [\text{video_start}, \text{video_start} + \text{pulse_duration}]$$

$$ALS_pulse(t) = 0 \text{ otherwise}$$

Pass criteria include:

visible analog rise
return to baseline
no clipping for full pulse duration
pulse present for Control, Target, and Override

Boundary:

ALS_pulse(t) belongs to u(t), not Y_topo(t).

5. StimulusFingerprint v1.8

StimulusFingerprint estimates the physical stimulus structure so physiological modules can be residualized against exogenous drive.

5.1 Visual luminance

For frame f with pixels $P_f(x,y)$ converted to grayscale intensity:

$$L_f = \text{mean}_{\{x,y\}}(\text{gray}(P_f(x,y)))$$

Resampled to time:

$$L(t) = \text{interpolate}(\{\text{frame_time_f}, L_f\})$$

5.2 Visual change / motion proxy

A simple frame-difference proxy:

$$V_{\text{change_f}} = \text{mean}_{\{x,y\}}(|\text{gray}(P_f) - \text{gray}(P_{f-1})|)$$

An optical-flow proxy may be substituted:

$$\text{Flow_f} = \text{mean}_{\{x,y\}}(\sqrt{u_{\text{flow}}(x,y)^2 + v_{\text{flow}}(x,y)^2})$$

5.3 Histogram delta

$$H_f = \text{normalized_histogram}(\text{gray}(P_f))$$

$$\text{HistDelta_f} = ||H_f - H_{f-1}||_1$$

5.4 Cut train

$$\text{Cut_f} = 1[\text{HistDelta_f} > \text{theta_cut}]$$

where theta_cut is calibrated from the distribution of frame changes.

5.5 Flash-event proxy

$$\text{Flash_f} = 1[|L_f - L_{f-1}| > \text{theta_flash}]$$

5.6 Audio RMS envelope

For audio waveform $a[n]$, sample rate fs , and window length N :

$$\text{RMS}_k = \sqrt{(1/N) * \sum_{n \text{ in window } k} a[n]^2}$$

$$\text{dBFS}_k = 20 * \log_{10}(\text{RMS}_k + \text{eps})$$

Envelope derivative:

$$\text{AudioDeriv}_k = \text{RMS}_k - \text{RMS}_{\{k-1\}}$$

5.7 Cue regressors

$\text{CueOn}(t) = 1$ if any cue visible at t , else 0

$\text{CueImpulse}(t) = \sum_i \delta(t - t_{i_start})$

$\text{CueValue}(t) = v_i$ during cue i , else 0

$\text{CuePhaseSin}(t) = \sin(2*\pi*t / T_{\text{cue_interval}})$

$\text{CuePhaseCos}(t) = \cos(2*\pi*t / T_{\text{cue_interval}})$

The cue frequency is:

$$f_{\text{cue}} = 1 / T_{\text{cue_interval}}$$

For $T_{\text{cue_interval}} = 3.0$ s:

$$f_{\text{cue}} = 0.333 \text{ Hz}$$

5.8 Dominant frequency map

For each stimulus regressor $u_m(t)$, compute a short-time Fourier transform:

$$U_m(f, \tau) = \text{STFT}[u_m(t)]$$

Dominant local frequency:

$$f_{m_star}(\tau) = \text{argmax}_f |U_m(f, \tau)|^2$$

5.9 Anchor stimulus vector

For anchor window W_j :

$$\text{StimVec}_j = [\text{mean}(L), \text{std}(L), \text{mean}(V_{\text{change}}), \text{cut_rate}, \text{flash_rate}, \\ \text{mean}(\text{AudioRMS}), \text{std}(\text{AudioRMS}), \text{cue_density}, f_{\text{star_audio}}, f_{\text{star_visual}}]_{W_j}$$

This vector feeds CET/EET and stimulus-side QC.

6. Core EEG spectral features

6.1 Filtering

For a band $[f1, f2]$, a band-limited signal is:

$$x_{\text{band},ch}(t) = \text{BandPass}(x_{\text{ch}}(t), f1, f2)$$

6.2 Welch power spectral density

For window W_t :

$$\text{PSD}_{\text{ch}}(f,t) = \text{WelchPSD}(x_{\text{ch}} \text{ over } W_t)$$

$$\text{BandPower}_{\text{ch},b}(t) = \text{integral}_{\{f \text{ in band } b\}} \text{PSD}_{\text{ch}}(f,t) df$$

$$\text{LogBandPower}_{\text{ch},b}(t) = \log(\text{BandPower}_{\text{ch},b}(t) + \text{eps})$$

6.3 ROI aggregation

For ROI R with channels ch in R :

$$\text{BandPower}_{R,b}(t) = \text{mean}_{\{ch \text{ in } R\}}(\text{LogBandPower}_{\text{ch},b}(t))$$

or robust median:

$$\text{BandPower}_{R,b}(t) = \text{median}_{\{ch \text{ in } R\}}(\text{LogBandPower}_{\text{ch},b}(t))$$

6.4 Phase and phase-locking value

Using the analytic signal:

$$z_{\text{ch},b}(t) = \text{Hilbert}(x_{\text{band},ch}(t))$$

$$\text{phase}_{\text{ch},b}(t) = \text{angle}(z_{\text{ch},b}(t))$$

Pairwise phase-locking:

$$\text{PLV}_{\{a,b\}}(W) = | (1/N) * \sum_{\{t \text{ in } W\}} \exp(i * (\text{phase}_a(t) - \text{phase}_b(t))) |$$

ROI PLV:

$$\text{PLV}_{\text{ROI}}(W) = \text{mean}_{\{\text{channel pairs in ROI}\}}(\text{PLV}_{\text{pair}}(W))$$

7. ArtifactScore

ArtifactScore is a veto/covariate layer, not a biological finding.

7.1 Peak-to-peak amplitude

$$\text{p2p}_{\text{ch}}(W) = \max_{\{t \text{ in } W\}}(x_{\text{ch}}(t)) - \min_{\{t \text{ in } W\}}(x_{\text{ch}}(t))$$

7.2 High-frequency sentinel

A generic muscle/line-noise sentinel:

$$\text{HF_ch}(W) = \log(\text{integral}_{\{45 \text{ Hz}\}^{\{55 \text{ Hz}\}}} \text{PSD_ch}(f,W) \, df + \text{eps})$$

Optional wider EMG sentinel:

$$\text{EMG_ch}(W) = \log(\text{integral}_{\{55 \text{ Hz}\}^{\{95 \text{ Hz}\}}} \text{PSD_ch}(f,W) \, df + \text{eps})$$

7.3 Ocular/frontal sentinel

For frontal channel group F:

$$\text{FpSentinel}(W) = \text{mean}_{\{\text{ch in } F\}}(z(\text{p2p_ch}(W)) + z(\text{HF_ch}(W))) / 2$$

7.4 Composite artifact score

Default operational form:

$$\begin{aligned} \text{ArtifactScore}(W) = & \text{mean}(\\ & \text{rz}(\text{p2p_all}(W)), \\ & \text{rz}(\text{HF_45_55}(W)), \\ & \text{rz}(\text{EMG_55_95}(W)), \\ & \text{rz}(\text{FpSentinel}(W)), \\ & \text{rz}(\text{line_noise_ratio}(W)), \\ & \text{rz}(\text{flatline_or_step_flag}(W)) \\ &) \end{aligned}$$

A stricter gate uses:

$$\text{ArtifactPass}(W) = 1[\text{ArtifactScore}(W) < \text{theta_artifact}]$$

For peak modules:

$$\text{QC}_j = \text{ArtifactPass}(W_{\text{hit},j}) * \text{TimingPass}_j * \text{ConfoundPass}_j$$

8. API_A v1: autonomic availability index

API_A is a proxy for regulated availability, not valence, clinical calm, or proof of parasympathetic healing.

8.1 HR and HRV features

Given R-R intervals RR_k in milliseconds:

$HR_k = 60000 / RR_k$
 $RMSSD = \sqrt{\text{mean} ((RR_{\{k+1\}} - RR_k)^2)}$
 $SDNN = \text{std}(RR_k)$
 $pNN50 = \text{count}(|RR_{\{k+1\}} - RR_k| > 50 \text{ ms}) / \text{count}(\text{pairs})$

HR slope over a window:

$HR_slope(W) = \text{slope of linear regression } HR(t) \sim t \text{ over } W$

8.2 API_A formula

Default operational form:

$API_A(t) = 0.30 * z(-HR_mean(t))$
 $+ 0.25 * z(RMSSD(t))$
 $+ 0.20 * z(SDNN(t))$
 $+ 0.15 * z(pNN50(t))$
 $+ 0.10 * z(-HR_slope(t))$
 $- 0.10 * z(RespInstability(t))$

If some HRV features are unavailable, weights should be renormalized over available features.

Interpretation:

higher API_A = lower HR / higher HRV / more regulated physiology, after respiration caution

9. GammaScalpel

GammaScalpel treats lower-gamma as an artifact-sensitive work-signal family, not a meaning biomarker.

9.1 Microband decomposition

Define 5 Hz bands:

$B_1 = 30\text{-}35 \text{ Hz}$
 $B_2 = 35\text{-}40 \text{ Hz}$
 $B_3 = 40\text{-}45 \text{ Hz}$
 $B_4 = 45\text{-}50 \text{ Hz}$ (more artifact-sensitive)
 $B_5 = 50\text{-}55 \text{ Hz}$ (line / sentinel caution)

For each ROI R and band b:

$\text{Gamma_R,b}(t) = \text{rz}(\text{LogBandPower_R,b}(t))$

9.2 MeaningGamma proxy

MeaningGamma is a weighted candidate signal. It should not be interpreted without artifact and stimulus controls.

$$\begin{aligned} \text{MeaningGamma}(t) = & \sum_b w_b * [\\ & a_{PT} * \text{Gamma_posterior_temporal}, b(t) \\ & + a_P * \text{Gamma_parietal}, b(t) \\ & + a_O * \text{Gamma_occipital}, b(t) \\ & - a_F * \text{Gamma_frontal_task}, b(t) \\ &] \\ & - \lambda_{art} * \text{ArtifactScore}(t) \\ & - \lambda_{vis} * \text{VisualDrive}(t) \\ & - \lambda_{cue} * \text{CueOn}(t) \end{aligned}$$

Typical interpretation:

posterior/temporal + parietal lower-gamma -> candidate semantic/scene recognition work

frontal/task gamma -> analytic extraction or task stance

visual/cue/artifact terms -> non-meaning penalties

9.3 TaskGamma proxy

$$\begin{aligned} \text{TaskGamma}(t) = & \sum_b w_b * [\\ & a_F * \text{Gamma_frontal}, b(t) \\ & + a_C * \text{Gamma_central}, b(t) \\ & + a_{PLV} * \text{PLV_fronto_central}, b(t) \\ &] \\ & - \lambda_{art} * \text{ArtifactScore}(t) \end{aligned}$$

TaskGamma is expected to rise during Contextual Override when the subject is doing the running-sum task.

9.4 GammaScalpel band decision

A band is considered interpretable only if:

$$\begin{aligned} \text{BandPass}_b = & 1[\text{Artifact}_b < \text{threshold}] \\ & * 1[\text{LineNoise}_b < \text{threshold}] \\ & * 1[\text{VisualDrive}_b \text{ not sufficient explanation}] \end{aligned}$$

Then:

$$\text{MeaningGamma_valid}(t) = \text{MeaningGamma}(t) * \text{BandPass}(t)$$

10. Temporal Semantic Proxy (TSP)

TSP estimates a candidate time of semantic recognition. It is not a ground-truth reading of meaning.

10.1 Components

$Gm(t)$ = MeaningGamma(t)

$PT(t)$ = posterior-temporal semantic-work proxy

$N(t)$ = NAST / absorption-transition support, if available

$K(t)$ = local KHT-topo support, if available

$A(t)$ = ArtifactScore(t)

$V(t)$ = VisualDrive(t)

$X(t)$ = TaskGamma(t)

10.2 TSP formula

$$\begin{aligned} TSP(t) = & w_G * rz(Gm(t)) \\ & + w_PT * rz(PT(t)) \\ & + w_N * rz(N(t)) \\ & + w_K * rz(K(t)) \\ & - w_X * rz(X(t)) \\ & - w_A * rz(A(t)) \\ & - w_V * rz(V(t)) \end{aligned}$$

In many runs, TSP is treated as:

$$TSP_z(t) = rz(TSP(t))$$

10.3 Event peak

For anchor window W_j :

$$t_peak,j = \operatorname{argmax}_{\{t \text{ in } W_j\}}(TSP_z(t))$$

$$TSP_peak,j = \max_{\{t \text{ in } W_j\}}(TSP_z(t))$$

A strict event should not rely only on TSP. TSP identifies candidate timing; MRED/A-MRED tests whether recognition and integration survive the gates.

11. Theta handoff / integration proxy

11.1 Theta feature

$$\Theta_R(t) = rz(\log \int_{4 \text{ Hz}}^{8 \text{ Hz}} PSD_R(f,t) df)$$

Use frontal-midline / frontocentral / parietal integration ROI as configured.

11.2 Post-event theta delta

For event time t_j :

$\text{Theta_pre},j = \text{mean}(\text{Theta_R}(t) \text{ over } [t_j - 10, t_j])$

$\text{Theta_post_0_10},j = \text{mean}(\text{Theta_R}(t) \text{ over } [t_j, t_j + 10])$

$\text{Theta_post_10_30},j = \text{mean}(\text{Theta_R}(t) \text{ over } [t_j + 10, t_j + 30])$

$\text{H_theta_0_10},j = \text{Theta_post_0_10},j - \text{Theta_pre},j$

$\text{H_theta_10_30},j = \text{Theta_post_10_30},j - \text{Theta_pre},j$

11.3 Handoff criterion

$\text{ThetaHandoff}_j = 1[\text{H_theta_10_30},j > \text{theta_handoff_threshold}]$

The delayed 10-30 s window is often more relevant than the immediate 0-10 s window when testing reflective integration rather than sensory surprise.

12. CandidateLocal_KHT-topo

CandidateLocal_KHT-topo is an event-level coupling model.

12.1 Human-scale state variables

E_t = observed work-signal proxy, e.g. MeaningGamma or TSP

Y_t = inferred human-translation state proxy, e.g. theta/topology/NIP state

u_t = exogenous stimulus regressors

A_t = artifact/confound regressors

12.2 Local reciprocal model

For local event window W_j :

$E_{\{t+\Delta\}} = a_E * E_t + \eta_{HT} * Y_t + b_E^T u_t + c_E * \text{Artifact}_t + \text{eps}_{E,t}$

$Y_{\{t+\Delta\}} = a_Y * Y_t + \zeta_{HT} * E_t + b_Y^T u_t + c_Y * \text{Artifact}_t + \text{eps}_{Y,t}$

where:

η_{HT} = $Y \rightarrow E$ coefficient

ζ_{HT} = $E \rightarrow Y$ coefficient

12.3 Local coupling coefficient

$K_{HT_topo},j = \sqrt{\text{abs}(\eta_{HT},j * \zeta_{HT},j)}$

$\text{LoopSign}_j = \text{sign}(\eta_{HT},j * \zeta_{HT},j)$

12.4 Event-lock rule

$KLock_j = 1[K_HT_topo,j \geq threshold_K \text{ or } percentile_K \geq threshold_percentile]$

$EventLock_j = KLock_j * ThetaHandoff_j * ArtifactPass_j * TimingPass_j$

K alone is not enough. Theta handoff alone is not enough.

12.5 Model-selection guard

For candidate models:

$M_full = \text{reciprocal } E \leftrightarrow Y$

$M_replay = \text{replay-only / no reciprocal hidden loop}$

$M_eta = \text{eta-only}$

$M_zeta = \text{zeta-only}$

$M_null = \text{no-coupling}$

$M_generic = \text{generic hidden oscillator}$

Use held-out prediction:

$CVWin_j = 1[CVError_full < \min(CVError_alternatives)]$

A strong candidate requires:

$EventLock_j * CVWin_j = 1$

13. MRED: Meaning Recognition / Encoding Dissociation

MRED separates meaning recognition from integration/carryover.

13.1 Meaning recognition score

$MR_j = w_TSP * TSP_peak,j$
 $+ w_G * MeaningGamma_peak,j$
 $+ w_K * KHT_topo_j$
 $+ w_N * NAST_j$
 $- w_A * ArtifactScore_j$
 $- w_V * VisualDrive_j$
 $- w_C * ConfoundBurden_j$

13.2 Encoding / integration score

$ENC_j = w_H * H_theta_10_30,j$
 $+ w_T * TopoShift_j$
 $+ w_E * EET_echo_j$
 $+ w_API * API_A_post,j$

```

+ w_B2 * Baseline2Echo_j
- w_A * ArtifactScore_post,j
- w_X * TaskGamma_post,j

```

13.3 MRED quadrant

Given thresholds θ_{MR} and θ_{ENC} :

```

if MR_j >=  $\theta_{MR}$  and ENC_j >=  $\theta_{ENC}$ :
    quadrant = MR_HIGH_ENC_HIGH
elif MR_j >=  $\theta_{MR}$  and ENC_j <  $\theta_{ENC}$ :
    quadrant = MR_HIGH_ENC_LOW
elif MR_j <  $\theta_{MR}$  and ENC_j >=  $\theta_{ENC}$ :
    quadrant = MR_LOW_ENC_HIGH
else:
    quadrant = MR_LOW_ENC_LOW

```

Interpretation:

MR_HIGH_ENC_HIGH = recognition plus integration candidate

MR_HIGH_ENC_LOW = recognition without clear integration; can occur with familiarity

MR_LOW_ENC_HIGH = possible nonsemantic update, task/cue/respiration/artifact caution

MR_LOW_ENC_LOW = no detected event under current criteria

14. A-MRED: anchor-locked primary endpoint

A-MRED is the compressed primary endpoint.

14.1 Strict gate

For a predeclared anchor j :

```

A_MRED_j =
    1[AnchorLocked_j = 1]
    * 1[MR_T,j >  $\theta_{MR}$ ]
    * 1[ENC_T,j >  $\theta_{ENC}$ ]
    * 1[MR_T,j > MR_C,j +  $\delta_{MR}$ ]
    * 1[MR_T,j > MR_O,j +  $\delta_{MR}$ ]
    * 1[ENC_T,j > ENC_C,j +  $\delta_{ENC}$ ]
    * 1[ENC_T,j > ENC_O,j +  $\delta_{ENC}$ ]
    * 1[QC_j = PASS]

```

where:

$QC_j = \text{ArtifactPass}_j * \text{TimingPass}_j * \text{ConfoundPass}_j * \text{CETRPass}_j$

14.2 Continuous backup score

$MREDScore_condition,j = \text{clip_plus}(MR_condition,j) * \text{clip_plus}(ENC_condition,j) * QC_condition,j$

$\Delta MRED_j = MREDScore_T,j - \max(MREDScore_C,j, MREDScore_O,j)$

14.3 Claim-grade modifiers

STRICT_CONFIRMATORY:

anchor locked before acquisition, frame verified, runner registered, ALS/timing pass.

RUNNER_REGISTERED_ESTIMATED:

anchor loaded before acquisition but estimated/not frame verified.

CONCEPTUALLY_PREDECLARED:

anchor existed in planning but not machine registered.

EXPLORATORY:

event found after physiology review.

15. MRED-Peak vs MRED-Resolution

This module separates acute meaning shock from delayed reflective closure.

15.1 MRED-Peak

MRED-Peak asks whether an anchor produces acute Target-specific recognition and delayed integration.

$\text{PeakEvidence}_j = \text{mean}(\begin{aligned} &z(MR_T,j), \\ &z(ENC_T,j), \\ &z(\Delta MRED_j), \\ &z(CII_margin_j), \\ &z(KHT_topo_j) \end{aligned})$

Strict peak pass:

$MRED_Peak_Pass_j = A_MRED_j \text{ or } BIT_j$

Soft peak candidate:

MRED_Peak_Candidate_j = 1[PeakEvidence_j > theta_peak] * 1[QC_j != FAIL]

15.2 MRED-Resolution

MRED-Resolution asks whether the event produces delayed reflective/regulatory recovery rather than a sharp event spike.

```
ResolutionEvidence_j = mean(  
    z(Baseline2Echo_j),  
    z(EET_j),  
    z(RDI_j),  
    z(TargetEchoedDuringBaseline2_report),  
    z(EmotionalAfterglow_T - EmotionalAfterglow_O),  
    z(API_A_B2 - API_A_B1)  
)
```

Resolution candidate:

```
MRED_Resolution_Candidate_j = 1[ResolutionEvidence_j > theta_resolution]  
    * 1[TargetSpecificity_j = 1]  
    * 1[QC_j != FAIL]
```

15.3 Run-level archetype

if max(PeakEvidence) high and ResolutionEvidence moderate:

 archetype = MRED_PEAK

elif ResolutionEvidence high and PeakEvidence moderate:

 archetype = MRED_RESOLUTION

elif MR high and ENC low:

 archetype = RECOGNITION_DOMINANT

else:

 archetype = UNRESOLVED

16. NIP: Narrative Immersion Proxy

NIP operationalizes attention plus resonance without claiming dopamine, oxytocin, or proprietary immersion biology.

16.1 Semantic attention component

```
A_sem(t) = clip_plus(  
    w_G * rz(MeaningGamma(t))  
    + w_T * rz(TSP(t))  
    + w_N * rz(NAST(t))  
    - w_V * rz(VisualDrive(t))
```

```

- w_A * rz(ArtifactScore(t))
- w_X * rz(TaskGamma(t))
)

```

16.2 Integration / resonance component

```

R_int(t) = clip_plus(
    w_H * rz(H_theta(t))
  + w_K * rz(KHT_topo(t))
  + w_TOPO * rz(TopoShift(t))
  + w_API * rz(API_A(t))
  - w_A * rz(ArtifactScore(t))
)

```

16.3 NIP density

With lag ell:

$$\text{NIP_density}(t) = A_sem(t)^{\alpha} * R_int(t + ell)^{\beta} * \text{Penalty}(t)$$

where:

$$\text{Penalty}(t) = \text{ArtifactPass}(t) * \text{ConfoundPenalty}(t) * \text{TimingPass}(t)$$

Default:

$\alpha = 1$

$\beta = 1$

17. BIT: Bivariate Immersion Threshold

BIT is an AND-gate, not an average.

```

BIT_j = 1[A_sem_j > theta_A]
      * 1[R_int_j > theta_R]
      * 1[Artifact_j < theta_artifact]
      * 1[Specificity_j = 1]

```

where:

```

Specificity_j = 1[Target_j > Control_j + delta]
              * 1[Target_j > Override_j + delta]

```

BIT fails when recognition is high but integration is absent:

MR_HIGH_ENC_LOW -> BIT = 0

This prevents a high attention spike from being mislabeled as full immersion.

18. CII: Continuous Immersion Index

CII measures average immersion density across an anchor window.

$$\text{CII_condition}(W_j) = (1 / |W_j|) * \int_{t \in W_j} \text{NIP_density_condition}(t) dt$$

Discrete implementation:

$$\text{CII_condition}(W_j) = \text{mean}_{\{t \in W_j\}}(\text{NIP_density_condition}(t))$$

Target-specific margin:

$$\text{CII_margin}_j = \text{CII_T},j - \max(\text{CII_C},j, \text{CII_O},j)$$

Duration normalization matters:

without division by $|W_j|$, long scenes automatically score higher.

19. IAQ / ITQ: Immersion Attenuation Quotient

IAQ estimates how much Contextual Override attenuated Target immersion.

$$\text{IAQ}_j = 1 - (\text{CII_O},j + \text{eps}) / (\text{CII_T},j + \text{eps})$$

Equivalent difference form:

$$\text{ITQ}_j = \text{CII_T},j - \text{CII_O},j$$

Interpretation:

IAQ near 1 = Override strongly attenuated the immersion proxy

IAQ near 0 = Override preserved a similar immersion proxy

IAQ below 0 = Override exceeded Target on the immersion proxy

Use IAQ in technical documents; reserve “theft” language for clearly labeled philosophical/cultural discussion.

20. TTI: Reception-Extraction Tradeoff

TTI estimates whether the same narrative was received or converted into a task object.

20.1 Budget principle

$$c_R R(t) + c_X X(t) + c_M M(t) + c_{\Gamma} \Gamma(t) + c_{RX} R(t)X(t) \leq B(t)$$

where:

$R(t)$ = receptive absorption

$X(t)$ = extractive / instrumental task stance

$M(t)$ = semantic payload

$\Gamma(t)$ = cognitive/metabolic exhaust proxy

$B(t)$ = finite processing budget

$c_{RX} R(t)X(t)$ = collision cost

20.2 Dynamic model

$$\begin{aligned} dR/dt = & \alpha_M M(t)(1-R) + \alpha_A A(t) \\ & - \beta_X X(t)R(t) - \beta_{\Gamma} \Gamma(t)R(t) - \lambda_R R(t) \end{aligned}$$

$$\begin{aligned} dX/dt = & \alpha_T T(t)(1-X) + \alpha_C C(t) \\ & - \beta_R R(t)X(t) - \lambda_X X(t) \end{aligned}$$

20.3 TTI composite

$$\begin{aligned} TTI = & w_1 * z(MR_T - MR_O) \\ & + w_2 * z(ENC_T - ENC_O) \\ & + w_3 * z(API_T - API_O) \\ & + w_4 * z(X_O - X_T) \\ & - w_5 * z(|Artifact_T - Artifact_O|) \\ & - w_6 * z(|VisualDrive_T - VisualDrive_O|) \end{aligned}$$

Default weights:

$$w_1 MR = 0.28$$

$$w_2 ENC = 0.24$$

$$w_3 API_A = 0.14$$

$$w_4 X\text{-load} = 0.22$$

$$w_5 \text{Artifact} = 0.08 \text{ penalty}$$

$$w_6 \text{Visual} = 0.04 \text{ penalty}$$

Positive TTI means Target preserved more reception/integration while Override carried more extraction/task load.

21. NUPI: Narrative Update Polarity Index

NUPI asks what kind of update meaning demanded.

21.1 Accommodative Load Index (ALI)

```
ALI = mean(  
  z(semantic_intensity),  
  z(Target_specificity),  
  z(complexity_perturbation),  
  z(TTI_positive_load),  
  z(primary_endpoint_pass_strength)  
)
```

Interpretation:

High ALI = model-expansion, awe, shock, destabilizing accommodation, or high update demand.

21.2 Regulatory Dividend Index (RDI)

```
RDI = mean(  
  z(Baseline2_regulation),  
  z(Baseline2_semantic_echo),  
  z(EET_afterstate_echo),  
  z(TargetEchoedDuringBaseline2_selfreport),  
  z(API_A_B2 - API_A_B1)  
)
```

Interpretation:

High RDI = reflective recovery, closure, parasympathetic regulation, after-state coherence.

21.3 NUPI score

NUPI = RDI - ALI

Classification:

NUPI > theta_positive and RDI high:

RESOLUTIVE_RECOVERY

ALI high and RDI moderate/high:

HIGH_LOAD_WITH_RECOVERY

ALI high and RDI low:

ACCOMMODATIVE_LOAD

MR high but ENC/RDI unclear:

RECOGNITION_DOMINANT_OR_UNRESOLVED

Boundary:

NUPI does not measure heat, ATP, clinical healing, morality, or literal thermodynamics.

22. NAST: Narrative Absorption State Transition

NAST estimates transition from idle/analytic/resting state into narrative absorption.

22.1 Alpha/idling proxy

$$\text{Alpha_R}(t) = \text{rz}(\log \int_{8 \text{ Hz}}^{12 \text{ Hz}} \text{PSD_R}(f,t) df)$$

Absorption is often expected to show alpha desynchronization relative to baseline, but this is not universal.

22.2 NAST proxy

$$\begin{aligned} \text{NAST}(t) = & w_{\text{TSP}} * \text{rz}(\text{TSP}(t)) \\ & + w_{\text{G}} * \text{rz}(\text{MeaningGamma}(t)) \\ & + w_{\text{H}} * \text{rz}(\text{ThetaIntegration}(t)) \\ & - w_{\text{A}} * \text{rz}(\text{Alpha}(t)) \\ & - w_{\text{X}} * \text{rz}(\text{TaskGamma}(t)) \\ & - w_{\text{art}} * \text{rz}(\text{ArtifactScore}(t)) \end{aligned}$$

22.3 Phase transition contrast

For transition from phase p to phase q:

$$\Delta \text{NAST}_{\{p \rightarrow q\}} = \text{mean}(\text{NAST over early } q) - \text{mean}(\text{NAST over late } p)$$

Example:

WASHOUT_1 -> TARGET
WASHOUT_2 -> OVERRIDE

Interpretation:

positive Target transition = narrative absorption candidate

positive Override transition with TaskGamma = analytic reorientation candidate

23. OCM025: Override Cue Microstate analysis

OCM025 analyzes cue-locked micro-windows at 0.25 s resolution.

23.1 Cue-relative bins

For cue onset t_i :

```

pre_i    = [t_i - 0.50, t_i)
digit_i  = [t_i, t_i + 0.85]
update_i = [t_i + 0.25, t_i + 1.25]
maint_i  = [t_i + 1.25, min(t_i + 3.0, t_{i+1}))

```

23.2 Digit recognition gamma

```

DR_i = mean(MeaningGamma or visual/digit gamma over digit_i)
      - mean(same feature over pre_i)

```

23.3 Working-memory update theta

```

WMU_i = mean(frontal_theta over update_i) - mean(frontal_theta over pre_i)

```

23.4 Maintenance theta

```

MAINT_i = mean(frontal_theta over maint_i) - mean(frontal_theta over pre_i)

```

23.5 OCM score

```

OCM_i = 0.35*z(DR_i)
        + 0.35*z(WMU_i)
        + 0.20*z(MAINT_i)
        - 0.10*z(Artifact_i)

```

OCM is Override-only unless explicitly used as a cue confound check.

24. RSM: Running Sum Microstate Model

RSM estimates arithmetic/cue burden during Override.

24.1 Running sum

```

S_i = S_{i-1} + v_i

```

24.2 Carry features

```

prior_units_i = S_{i-1} mod 10
carry_required_i = 1[prior_units_i + v_i >= 10]
high_carry_load_i = max(0, prior_units_i + v_i - 9)
hard9_i = 1[v_i = 9 and prior_units_i in {7,8,9}]

```

24.3 Arithmetic Compute Load

```

ACL_i = beta1*z(log(1 + S_{i-1}))
        + beta2*carry_required_i
        + beta3*z(high_carry_load_i)
        + beta4*hard9_i
        + beta5*z(fatigue_or_elapsed_time_i)

```

24.4 Compute-stall score

$$\begin{aligned}\text{ComputeStall}_i &= 0.35 * z(\text{WMU}_i) \\ &+ 0.30 * z(\text{MAINT}_i) \\ &+ 0.25 * z(\text{T_close}_i) \\ &+ 0.10 * \text{OpenLoop}_i \\ &- 0.20 * z_{\text{positive}}(\text{Artifact}_i)\end{aligned}$$

where:

T_close_i = time until theta/update loop returns below closure threshold

$\text{OpenLoop}_i = 1[\text{T_close}_i > t_{\{i+1\}} - \text{safety_buffer}]$

24.5 Latent guess-risk

$$\begin{aligned}\text{GuessRisk}_i &= \text{sigmoid}(\\ &\quad \text{gamma0} \\ &+ \text{gamma1} * \text{ArithmeticStall}_i \\ &+ \text{gamma2} * \text{DeadlinePressure}_i \\ &+ \text{gamma3} * \text{ACL}_i \\ &+ \text{gamma4} * \text{CueVisibilityBurden}_i \\ &+ \text{gamma5} * \text{PriorUncertainty}_i \\ &- \text{gamma6} * \text{FinalSumConfidence} \\ &)\end{aligned}$$

This is a latent proxy, not proof that the subject guessed.

25. CVB: Cue Visibility / Legibility Burden

CVB estimates cue-reading strain.

$$\begin{aligned}\text{CVB}_i &= 0.45 * z(\text{HoldBurden}_i) \\ &- 0.25 * z(\text{DigitRecognition}_i) \\ &+ 0.15 * z(\text{VisualSearch}_i) \\ &+ 0.15 * z(1 - \text{CueContrast}_i) \\ &+ 0.15 * z(\text{CueBlurOrSmallness_report}) \\ &- 0.10 * z(\text{Artifact}_i)\end{aligned}$$

For small/blurred cues, contrast may be high while legibility remains poor. CVB should therefore not rely only on luminance contrast.

26. SquintProxy

SquintProxy estimates frontal visual-strain artifacts. It cannot confirm squinting.

```
SquintProxy_i = 0.30*z(Fp1_HF_i)
               + 0.25*z(Fp1_p2p_i)
               + 0.20*z(frontal_EMG_proxy_i)
               + 0.15*z(EyeStrainOrSquint_report_i)
               + 0.10*z(CueBlurOrSmallness_report_i)
               - 0.20*z(BlinkTemplate_i)
```

Interpretation:

high SquintProxy = possible visual strain / ocular or forehead artifact
not confirmed squint

EOG or eye tracking is required for confirmation.

27. Confound registry

Confound reports are covariates/veto aids.

27.1 Branch confound score

```
ConfoundBurden_branch = mean(
    AudioVideoSyncProblem,
    AudioComprehensionDifficulty,
    ExternalNoiseIntrusion,
    SpeakerVolumeDifficulty,
    CueBlurOrSmallness,
    EyeStrainOrSquint
) / 9
```

27.2 Gate

```
ConfoundPass_branch = 1[ConfoundBurden_branch < theta_confound]
```

or claim-grade modifier:

```
if ConfoundBurden_branch >= theta_caution:
    claim_level = claim_level + "_CONFOUND_CAUTION"
```

Confounds should lower claim strength, not create positive evidence.

28. CET: Cinematic Entrainment Tracking

CET quantifies how much EEG tracks exogenous stimulus rhythm.

28.1 Stimulus transform

$$U_m(f, \tau) = \text{STFT}[u_m(t)]$$

where u_m may be audio envelope, luminance, visual change, cut train, cue train, etc.

28.2 EEG transform

$$X_{ch}(f, \tau) = \text{STFT}[x_{ch}(t)]$$

28.3 Coherence

$$\text{Coh}_{\{u,x\}}(f, \tau) = |S_{\{u,x\}}(f, \tau)|^2 / (S_{\{u,u\}}(f, \tau) * S_{\{x,x\}}(f, \tau) + \epsilon)$$

28.4 Phase lag

$$\phi_{\{u,x\}}(f, \tau) = \text{angle}(S_{\{u,x\}}(f, \tau))$$

$$\text{lag}_{\{u,x\}}(f, \tau) = \phi_{\{u,x\}}(f, \tau) / (2\pi f)$$

Caution:

Stable lag can mean good tracking, not stalled biological momentum.

29. CET-R: residualization against exogenous drive

CET-R asks whether MRED/NIP/TTI survives after modeling stimulus rhythm.

29.1 Regression model

For feature $Y(t)$ such as NIP density, MeaningGamma, TSP, MR, or ENC:

$$Y(t) = \beta_0 + \beta_u^T U(t) + \beta_{art} * \text{Artifact}(t) + \epsilon(t)$$

where:

$$U(t) = [\text{audio_env}, \text{luminance}, \text{visual_change}, \text{optical_flow}, \text{cut_train}, \text{flash_train}, \text{cue_on}, \text{cue_value}, \text{cue_phase_sin}, \text{cue_phase_cos}, \text{ALS_pulse}]$$

Residual:

$$Y_{\text{resid}}(t) = Y(t) - \hat{Y}(t)$$

29.2 Blocked cross-validation

Split time into blocks, fit on training blocks, evaluate held-out blocks:

$$R2_blocked = 1 - \text{sum}((Y_test - Y_hat_test)^2) / \text{sum}((Y_test - \text{mean}(Y_train))^2)$$

If blocked-CV is poor or negative, the stimulus regressors do not generalize as an explanation.

29.3 Residualized CII

$$CII_resid(W_j) = \text{mean}_{\{t \text{ in } W_j\}}(\text{clip_plus}(Y_resid(t)))$$

A robust Target effect should remain Target-dominant after residualization.

30. EET: Endogenous Echo Tracking

EET asks whether a stimulus-anchor state vector resembles later washout/Baseline2 physiology.

30.1 State vector

For anchor j:

$$\text{StateVec_anchor},j = [\text{MR}, \text{ENC}, \text{NIP}, \text{TSP}, \text{MeaningGamma}, \text{Theta}, \text{Alpha}, \text{API_A}, \text{HRV}, \text{Resp}, \dots]_{Wj}$$

For after-state window q:

$$\text{StateVec_after},q = [\text{same features}]_{Wq}$$

30.2 Cosine similarity

$$\begin{aligned} \text{EET}_{\{j,q\}} &= \text{cosine}(\text{StateVec_anchor},j, \text{StateVec_after},q) \\ &= \text{dot}(V_j, V_q) / (||V_j|| * ||V_q|| + \text{eps}) \end{aligned}$$

30.3 Difference from control echo

$$\text{DeltaEET}_j = \text{EET}_{\{\text{Target anchor } j \rightarrow B2\}} - \text{EET}_{\{\text{Control matched window } j \rightarrow B2\}}$$

EET is state-vector resemblance. It is not proof of memory replay.

31. MRED-ITP: Information-Thermodynamic Proxy layer

MRED-ITP tests complexity perturbation/settlement and ocular release around MRED events.

31.1 ACG: Algorithmic Complexity Gate

Given symbolic sequence S_t derived from cleaned EEG window W_t :

$$C_LZ(t) = LZC(S_t) / E[LZC(\text{shuffle}(S_t))]$$

For anchor j:

$C_{pre,j} = \text{mean}(C_{LZ} \text{ over } [a_j - 15, a_j])$
 $C_{peak,j} = \text{mean}(C_{LZ} \text{ over } [a_j, a_j + 5])$
 $C_{post,j} = \text{mean}(C_{LZ} \text{ over } [a_j + 10, a_j + 30])$

$\Delta C_{strike,j} = C_{peak,j} - C_{pre,j}$
 $\Delta C_{settle,j} = C_{post,j} - C_{peak,j}$
 $\Delta C_{baseline,j} = C_{post,j} - C_{pre,j}$

ACG flag:

$ACG_j = 1[\Delta C_{strike,j} > \theta_{up}]$
 $* 1[\Delta C_{settle,j} < -\theta_{down}]$
 $* 1[\text{Artifact}_j < \theta_{artifact}]$

31.2 CSI: Complexity Settlement Index

$CSI_j = z(\Delta C_{strike,j}) - z(\Delta C_{post_minus_peak,j})$

where $\Delta C_{post_minus_peak}$ is positive when post remains above peak. A high CSI means perturbation followed by settlement.

31.3 OCU: Ocular-Cognitive Unloading

Blink detection proxy:

$\text{blink}_i = 1[\text{Fp1_p2p}_i > \theta_{p2p}]$
 and $\text{Fp1_lowfreq}_i > \theta_{lf}$
 and $\text{duration}_i \text{ in } [100, 500] \text{ ms}$

Blink rates:

$BR_{pre,j} = \text{count}(\text{blinks in } [a_j - 30, a_j - 5]) / 25$
 $BR_{hold,j} = \text{count}(\text{blinks in } [a_j - 5, a_j + 5]) / 10$
 $BR_{release,j} = \text{count}(\text{blinks in } [a_j + 5, a_j + 20]) / 15$

Suppression and release:

$\text{BlinkSuppression}_j = z(BR_{baseline} - BR_{hold,j})$
 $\text{BlinkRelease}_j = z(BR_{release,j} - BR_{hold,j})$

OCU index:

$OCU_j = 0.45 * z(\text{BlinkSuppression}_j)$
 $+ 0.45 * z(\text{BlinkRelease}_j)$
 $+ 0.10 * z(\text{EventBoundary}_j)$
 $- 0.25 * z_{\text{positive}}(\text{Artifact}_j)$

Strict OCU flag:

$$\begin{aligned} \text{OCUFlag}_j &= 1[\text{BlinkSuppression}_j > \text{theta}_S] \\ &\quad * 1[\text{BlinkRelease}_j > \text{theta}_R] \\ &\quad * 1[\text{Artifact}_j < \text{theta}_{\text{artifact}}] \end{aligned}$$

OCU is not proof of memory encoding.

32. LSO / SPM: Subtitle Override and Subtitle Phase Mapping

This is optional and not part of default PRAYCG2.0 unless selected.

32.1 Subtitle line representation

$$\text{Sub}_i = (\text{t}_{\text{on},i}, \text{t}_{\text{off},i}, \text{text}_i, \text{bbox}_i)$$

32.2 Lexical Extraction Cost

$$\begin{aligned} \text{CharsPerSec}_i &= N_{\text{chars},i} / (\text{t}_{\text{off},i} - \text{t}_{\text{on},i}) \\ \text{WordsPerSec}_i &= N_{\text{words},i} / (\text{t}_{\text{off},i} - \text{t}_{\text{on},i}) \\ \text{LineLoad}_i &= N_{\text{lines},i} \end{aligned}$$

$$\begin{aligned} \text{LEC}_i &= w_c * z(\text{CharsPerSec}_i) \\ &\quad + w_w * z(\text{WordsPerSec}_i) \\ &\quad + w_l * z(\text{LineLoad}_i) \\ &\quad + w_s * z(\text{SegmentationDifficulty}_i) \\ &\quad + w_g * z(\text{GazeRevisits}_i) \end{aligned}$$

32.3 Subtitle semantic timing

If gaze is available:

$$\text{t}_{\text{read},i} = \text{first time gaze completes subtitle bbox traversal}$$

If gaze is unavailable:

$$\text{t}_{\text{semantic},i} = \arg\max_{\{t \in [\text{t}_{\text{on},i} - 2, \text{t}_{\text{off},i} + 1]\}} \text{TSP}(t)$$

Phase shift relative to audio:

$$\phi_i = \text{t}_{\text{read_or_semantic},i} - \text{t}_{\text{audio_anchor},i}$$

32.4 Subtitle MRED

$$\begin{aligned} \text{MR}_{\text{sub},j} &= 0.30 * z(\text{MeaningGamma}_j) \\ &\quad + 0.30 * z(\text{TSP}_j) \\ &\quad + 0.20 * z(\text{KHT}_{\text{topo},j}) \\ &\quad + 0.20 * z(\text{LexicalRecognition}_j) \\ &\quad - 0.20 * z(\text{Artifact}_j) \end{aligned}$$

$$\begin{aligned} \text{ENC_sub,j} = & 0.55 * z(\text{H_theta_j}) \\ & + 0.15 * z(\text{API_A_j}) \\ & + 0.15 * z(\text{Novelty_j}) \\ & - 0.20 * z(\text{Familiarity_j}) \\ & - 0.25 * z(\text{LEC_j}) \\ & - 0.20 * z(\text{TaskGamma_j}) \\ & - 0.15 * z(\text{Artifact_j}) \end{aligned}$$

Subtitle choke:

$$\begin{aligned} \text{SubtitleChoke_j} = & 1[\text{MR_sub,j} > \text{theta_MR}] \\ & * 1[\text{ENC_sub,j} < \text{theta_ENC}] \\ & * 1[\text{LEC_j} > \text{theta_LEC}] \end{aligned}$$

33. Topo-OSM Network State

Topo-OSM is the human-scale interpretation layer for network-state topology.

33.1 Human-scale latent state

$$\begin{aligned} Y_{\text{topo}}(t) = & \text{Phi}[\\ & A_{\text{theta}}(t), \\ & A_{\text{gamma}}(t), \\ & A_{\text{alpha}}(t), \\ & \text{TSP}(t), \\ & \text{API_A}(t), \\ & \text{Resp}(t), \\ & \text{Artifact}(t), \\ & u(t) \\ &] \end{aligned}$$

This is not cellular Y_{cell} and not K_{OSM} .

33.2 State vector implementation

$$\text{TopoState}(t) = [\text{ThetaTopology}(t), \text{GammaTopology}(t), \text{AlphaState}(t), \text{TSP}(t), \text{NIP}(t), \text{API_A}(t), \text{Resp}(t), \text{Artifact}(t)]$$

33.3 Topological shift

For pre/post windows:

$$\text{TopoShift_j} = \text{distance}(\text{mean}(\text{TopoState over } W_{\text{post,j}}), \text{mean}(\text{TopoState over } W_{\text{pre,j}}))$$

Possible distances:

Euclidean: $\|V_{\text{post}} - V_{\text{pre}}\|_2$

Cosine: $1 - \text{cosine}(V_{\text{post}}, V_{\text{pre}})$

Mahalanobis: $\sqrt{(V_{\text{post}} - V_{\text{pre}})^T \Sigma^{-1} (V_{\text{post}} - V_{\text{pre}})}$

33.4 Persistence

$\text{TopoPersistence}_j = \text{mean}_{\{q \text{ in post windows}\}}(\text{cosine}(V_{\text{anchor},j}, V_q))$

Topo-OSM does not prove memory. It provides a human-scale state-topology proxy.

34. Baseline 1 vs Baseline 2

34.1 Feature delta

For feature F:

$\text{DeltaF}_{B2_B1} = \text{mean}(F \text{ over } B2) - \text{mean}(F \text{ over } B1)$

34.2 Reflective-state index

$\text{ReflectiveIndex} = \text{mean}(\text{z}(\text{DeltaTSP}_{B2_B1}), \text{z}(\text{DeltaNIP}_{B2_B1}), \text{z}(\text{DeltaAPI}_A_{B2_B1}), \text{z}(-\text{DeltaTaskGamma}_{B2_B1}), \text{z}(-\text{DeltaArtifact}_{B2_B1}))$

34.3 Regulation index

$\text{Baseline2Regulation} = \text{mean}(\text{z}(-\text{DeltaHR}_{B2_B1}), \text{z}(\text{DeltaRMSSD}_{B2_B1}), \text{z}(\text{DeltaSDNN}_{B2_B1}), \text{z}(\text{DeltaPNN50}_{B2_B1}), \text{z}(-\text{DeltaRespInstability}_{B2_B1}))$

These feed MRED-Resolution and NUPI.

35. SelfReport20 parser

Self-report is an independent evidence stream, not proof.

35.1 Branch core report vector

Report_branch = [Meaning, Absorption, EmotionalAfterglow, StoryActiveWashout,
TaskExtractionLoad, ConfoundBurden]

35.2 Final report vector

Report_final = [OverrideReducedReception, StoryBrokeThroughOverride,
TargetEchoedDuringBaseline2, Familiarity, NewMeaningToday,
CurrentLifeResonance]

35.3 Model relation

A generic mixed model:

Report_{i,k} = alpha
+ beta_MR * MR_{i,k}
+ beta_ENC * ENC_{i,k}
+ beta_NIP * NIP_{i,k}
+ beta_API * API_A_{i,k}
- beta_X * TaskGamma_{i,k}
- beta_A * Artifact_{i,k}
+ u_subject
+ eps_{i,k}

A positive correspondence is not proof of internal state; contradiction is a warning.

36. Visualizer formulas

The visualizer does not create scientific endpoints. It audits synchronization and module outputs.

36.1 Robust axis scaling

For visible panel window W_vis:

y_min = percentile(y over W_vis, 5)
y_max = percentile(y over W_vis, 95)

or z-score mode:

y_plot = clip(rz(y), -z_clip, z_clip)

36.2 Missing-signal gate

```
UsableFeature = 1[non_null_count >= N_min]
    * 1[std(y) > eps]
    * 1[unique_count > 1]
```

If UsableFeature = 0, the panel should display “signal unavailable” rather than a fake flat line.

36.3 Event card display

For event time e_k :

```
ShowEventCard_k(t) = 1[ |t - e_k| < card_window ]
```

37. Offline interpretive report generator

The offline interpreter is deterministic and rule-based.

37.1 Table detection

```
ModulePresent_m = 1[expected_table_m exists]
```

37.2 Rule evaluation

Example:

```
if A_MRED_pass_count > 0 and claim_level strict:
    statement = "A-MRED positive under current strict gate."
elif A_MRED_pass_count > 0 and timing_caution:
    statement = "A-MRED pilot-positive with timing caution."
else:
    statement = "No strict A-MRED pass detected."
```

37.3 Boundary

The offline interpreter explains tables.

It does not create new evidence.

It does not replace expert review.

38. Opto-PING / Rung 0 synthetic model

Opto-PING is synthetic identifiability and model-comparison support. It does not prove human biology.

38.1 Orthodox PING skeleton

$E(t) \leftrightarrow I(t)$

where:

$E(t)$ = excitatory population activity

$I(t)$ = inhibitory population activity

38.2 Hidden reciprocal model

$E_{t+1} = a_E E_t + \eta_E Y_t + b_E u_t + \epsilon_{E,t}$

$Y_{t+1} = a_Y Y_t + \zeta_E E_t + b_Y u_t + \epsilon_{Y,t}$

38.3 Loop coefficient

$K_{\text{loop}} = \eta_E * \zeta_E$

$K = \sqrt{\text{abs}(\eta_E * \zeta_E)}$

K is more stable than separately interpreting η and ζ .

38.4 Joint negative log likelihood

For conditions $c = 1..C$ and Kalman innovations $v_k^{(c)}$ with covariance $S_k^{(c)}$:

$\text{NLL}(\theta) = \sum_c \sum_k [\log |S_k^{(c)}| + v_k^{(c)T} * \text{inv}(S_k^{(c)}) * v_k^{(c)}]$

where:

$\theta = [\eta_E, \zeta_E, \text{process noise}, \text{observation noise}, \dots]$

38.5 Model-selection gate

$\text{Pass_Rung0} = 1[K_{\text{error}} \leq \text{threshold}]$

$\quad * 1[\text{FullModel_CV_WinRate} \geq \text{threshold}]$

$\quad * 1[\text{NullAlternatives do not match full model}]$

Boundary:

Synthetic identifiability only.

No biological or human EEG mechanism claim.

39. Module interaction map

A single event can pass through the suite as:

1. MediaPrep creates stimulus + cue + anchor scaffold.
2. Runner records which files were used and writes Stasis markers.
3. Feature extractor computes EEG/autonomic/stimulus features.

4. ArtifactScore and confound registry define veto/caution layers.
 5. TSP/GammaScalpel detect candidate recognition timing.
 6. Theta/Topo/after-state features estimate integration.
 7. MRED separates recognition from integration.
 8. A-MRED applies the strict primary endpoint gate.
 9. NIP/CII summarizes continuous immersion density.
 10. TTI compares reception vs extraction.
 11. NUPI classifies update polarity.
 12. CET-R tests whether stimulus rhythm explains the effect.
 13. EET/MRED-ITP/OCU/ACG provide exploratory convergence.
 14. Visualizer and offline interpreter explain the result without adding evidence.
-

40. Example event derivation

For a predeclared Target anchor j:

- Step 1: compute MeaningGamma_j and TSP_j.
- Step 2: compute MR_{T,j}.
- Step 3: compute H_theta_10_30,j, TopoShift_j, API_A_post,j.
- Step 4: compute ENC_{T,j}.
- Step 5: repeat matched windows for Control and Override.
- Step 6: compute ArtifactPass, ConfoundPass, TimingPass, CETRPass.
- Step 7: compute A_MRED_j.
- Step 8: compute CII_{T/C/O} and IAQ.
- Step 9: compute PeakEvidence and ResolutionEvidence.
- Step 10: write visual overlay and offline interpretation.

Mathematically:

$$A_MRED_j = \text{Gate}(MR_T,j, ENC_T,j, MR_C,j, ENC_C,j, MR_O,j, ENC_O,j, QC_j)$$

$$CII_margin_j = CII_T,j - \max(CII_C,j, CII_O,j)$$

$$PeakEvidence_j = f(MR_T,j, ENC_T,j, CII_margin_j, KHT_topo_j, QC_j)$$

$$ResolutionEvidence_j = f(B2_echo_j, EET_j, RDI_j, self_report_echo_j, API_recovery_j)$$

41. Formula safety notes

41.1 The suite contains operational indices

These formulas are engineered proxies. They are not direct measurements of hidden essence.

MeaningGamma != meaning

TSP != true semantic time

KHT-topo != K_OSM

EET != replay proof

OCU != memory dump proof

NUPI != heat or ATP

TTI != moral theft

A-MRED != consciousness proof

41.2 Positive result language

Use:

Target-specific recognition-plus-integration candidate.

A-MRED pass under current gates.

MRED-Peak candidate.

MRED-Resolution candidate.

Resolvable recovery profile.

Accommodative-load profile.

Avoid:

proved consciousness

proved memory formation

proved OSM biology

proved the soul

proved thermodynamic meaning

41.3 Negative result language

Use:

No strict pass under current operational criteria.

Recognition may have occurred without detectable integration.

Timing or artifact caveats prevent strict interpretation.

Avoid:

nothing happened

the scene was meaningless

the subject felt nothing

42. Minimal confirmatory endpoint recommendation

For a future group or preregistered study, the primary endpoint should be one of:

A-MRED pass proportion across locked anchors

DeltaMRED continuous score across locked anchors

MRED-Peak vs MRED-Resolution classification across locked anchors

The confirmatory statistical model could be:

$A_MRED_pass \sim Condition + (1 | Subject) + (1 | Stimulus) + (1 | Anchor)$

or continuous:

$\Delta MRED \sim Condition + Familiarity + Artifact + CETR + (1 | Subject) + (1 | Stimulus) + (1 | Anchor)$

Prediction:

Target > Control

Target > Override

The rest of the suite should be treated as secondary or exploratory unless preregistered as primary.

43. Appendix A: recommended output files by module

Core feature extraction:

*_time_resolved_feature_frame.csv

*_phase_feature_summary.csv

A-MRED:

amred_anchor_endpoint_table.csv

amred_primary_endpoint_summary.csv

amred_visual_overlay.csv

MRED-Peak/Resolution:

mred_peak_resolution_anchor_table.csv

mred_peak_resolution_run_summary.csv

top_mred_peak_candidates.csv

top_mred_resolution_candidates.csv

mred_peak_resolution_visual_overlay.csv

MRED / KHT:

- mred_event_table.csv
- candidate_local_kht_topo_mred_event_table.csv
- candidate_local_kht_visual_overlay.csv

NIP / CET / EET:

- nip_component_timeseries.csv
- bit_event_table.csv
- cii_anchor_integrals.csv
- iaq_target_override_table.csv
- cet_residualization_model_summary.csv
- eet_endogenous_echo_tracking.csv

TTI / NUPI:

- tti_global_summary.csv
- tti_anchor_deltas.csv
- nupi_run_summary.csv
- nupi_anchor_polarity_table.csv

OCM/RSM/CVB/Squint:

- ocm025_rsm_cvb_squint_cue_epoch_table.csv
- rsm_correlation_summary.csv
- confound_registry.csv

MRED-ITP:

- lz_complexity_timeseries.csv
- acg_event_table.csv
- blink_event_table.csv
- ocu_event_table.csv
- mred_itp_anchor_summary.csv

Visualizer:

- *_features_used.csv
- *_events_used.csv
- *_feature_diagnostics.csv
- *_event_time_diagnostics.csv
- *_render_report.json

Offline interpreter:

- reports/offline_interpretation/offline_interpretive_report.md
- reports/offline_interpretation/offline_interpretive_report.txt
- reports/offline_interpretation/offline_interpretive_report_summary.json

44. Appendix B: file provenance rule

MediaPrep:

stimulus videos, cue schedules, draft/locked anchor scaffolds, stimulus fingerprints.

Protocol runner:

event markers, media file hashes, run configuration, self-reports, confound reports.

Master Comprehensive Suite:

feature tables, module outputs, event tables, endpoint summaries.

Visualizer:

synchronized MP4 and sidecar tables.

Offline interpreter:

deterministic text interpretation of existing module outputs.

Clean rule:

If a file describes what was shown, it probably comes from MediaPrep or the runner.

If a file describes physiology over time, it comes from the Master Comprehensive Suite.

If a file explains what the outputs mean, it comes from the offline interpreter or report generator.

45. Appendix C: compact formula catalog

$$rz(x) = (x - \text{median_ref}) / (1.4826 * \text{MAD_ref} + \text{eps})$$
$$\text{PLV} = |\text{mean}(\exp(i * (\text{phase_a} - \text{phase_b})))|$$
$$\text{ArtifactScore} = \text{mean}(rz(p2p), rz(HF_{45_55}), rz(EMG), rz(FpSentinel), rz(line_noise), \text{step_flag})$$
$$\text{API_A} = .30 * z(-HR) + .25 * z(RMSSD) + .20 * z(SDNN) + .15 * z(pNN50) + .10 * z(-HR_slope) - .10 * z(\text{RespInstability})$$
$$\text{TSP} = w_G * \text{MeaningGamma} + w_PT * \text{PosteriorTemporal} + w_N * \text{NAST} + w_K * \text{KHT} - w_X * \text{TaskGamma} - w_A * \text{Artifact} - w_V * \text{VisualDrive}$$
$$\text{KHT_topo} = \text{sqrt}(\text{abs}(\text{eta_HT} * \text{zeta_HT}))$$
$$\text{MR} = w_TSP * \text{TSP} + w_G * \text{MeaningGamma} + w_K * \text{KHT} + w_N * \text{NAST} - \text{penalties}$$
$$\text{ENC} = w_H * \text{ThetaHandoff} + w_T * \text{TopoShift} + w_E * \text{EET} + w_API * \text{API_A} + w_B2 * \text{B2Echo} -$$

penalties

$$A_MRED = 1[MR_T > th] * 1[ENC_T > th] * 1[T > C] * 1[T > O] * 1[QC = PASS]$$

$$NIP_density = clip_plus(A_sem)^{\alpha} * clip_plus(R_int_lag)^{\beta} * Penalty$$

$$CII = \text{mean}(NIP_density \text{ over anchor window})$$

$$IAQ = 1 - (CII_Override + \epsilon) / (CII_Target + \epsilon)$$

$$TTI = w1 * z(MR_T - MR_O) + w2 * z(ENC_T - ENC_O) + w3 * z(API_T - API_O) + w4 * z(X_O - X_T) - \text{penalties}$$

$$NUPI = RDI - ALI$$

$$\text{CET residual: } Y_resid = Y - (\beta_0 + \beta_u^T U + \beta_{art} * \text{Artifact})$$

$$EET = \cosine(\text{StateVec_anchor}, \text{StateVec_after})$$

$$LZC_norm = LZC(S) / E[LZC(\text{shuffle}(S))]$$

$$OCU = .45 * z(\text{BlinkSuppression}) + .45 * z(\text{BlinkRelease}) + .10 * z(\text{EventBoundary}) - .25 * z(\text{Artifact})$$

$$LEC = w_c * z(\text{chars/sec}) + w_w * z(\text{words/sec}) + w_l * z(\text{line_count}) + w_s * z(\text{segmentation}) + w_g * z(\text{gaze_revisits})$$

$$\text{OptoPING } K = \sqrt{\text{abs}(\eta_E * \zeta_E)}$$

46. Final statement

The Master Comprehensive Analysis Suite is best understood as a layered falsification machine. Its most useful result is not that every module produces a positive number. Its most useful result is that the same event can be tested against timing, artifact, sensory entrainment, task extraction, self-report, autonomics, and state persistence before the language of meaning is allowed into the report.

The formulas in this document give each layer a clear mathematical address. They should be treated as auditable operational definitions, not as metaphysical declarations.

47. v1.1 update: phase-scrambled Control as sensory-entrainment guard

47.1 Why this update exists

The phase-scrambled branch should not be described as physiologically meaningless. It is a meaning-damaged sensory-control condition. It can still drive visual entrainment, auditory entrainment, boredom, confusion, artifact, task-like scanning, startle, or low-level affect. Its job is not to be dead. Its job is to preserve enough low-level stimulus structure that Target-specific effects must survive a credible sensory-entrainment alternative.

The updated control definition is:

```
TargetVideo = cue_embed(clean_master)
OverrideVideo = TargetVideo
ControlVideo = phase_scramble(TargetVideo)
ControlAudio = speech_shaped_noise_envelope(TargetAudio)
```

The control branch can be modeled as:

$$\text{Control}(t) = \text{SensoryDrive}(t) + \text{CueEnergy}(t) + \text{ArtifactRisk}(t) - \text{RecognizableNarrative}(t)$$

Thus the serious endpoint is not:

$$\text{Control} = 0$$

but:

$$\text{Target} > \text{Control}$$

after artifact, stimulus-fingerprint, and CET-R gates.

47.2 Formal sensory-entrainment model

Let the exogenous stimulus vector be:

$$\mathbf{u}(t) = [\text{audio_env}(t), \text{luminance}(t), \text{visual_change}(t), \text{optical_flow}(t), \text{cut_train}(t), \text{cue_train}(t), \text{ALS_pulse}(t)]$$

A sensory-tracking prediction for a physiological feature is:

$$Y_{\text{feature}}(t) = \beta_0 + \sum_m \beta_m u_m(t - \tau_m) + \beta_{\text{art}} \text{Artifact}(t) + \epsilon(t)$$

The residualized feature is:

$$Y_{\text{resid}}(t) = \epsilon(t)$$

A Target-specific meaning candidate must remain after this operation:

$$\text{MeaningCandidate}(t) = f(Y_{\text{resid}}(t), D_{\text{gate}}(t), MR(t), ENC(t))$$

47.3 Control validity gates

The phase-scrambled branch remains required for serious PRAYCG. It can be skipped only for engineering tests or PRAYCG-lite demonstrations. Confirmation-grade interpretation requires:

ControlAudioQC = PASS
ControlVisualQC = PASS
ControlGeneratedFromCueEmbeddedTarget = TRUE
TargetOverrideHashMatch = TRUE
CET_R_available = TRUE

If Control is absent:

claim_grade = PRAYCG_LITE_NO_SENSORY_CONTROL

If Control audio remains recognizable:

claim_grade = PILOT_ONLY_CONTROL_AUDIO_FAIL

48. v1.1 update: DGA - Decoder Gate Availability

48.1 Rationale

DGA formalizes a dissociation exposed by the Eternal Sunshine run: semantic clarity and emotional integration can separate across branches. A Target can be emotionally absorbing despite degraded audio access; an Override can provide clearer semantic access while still imposing analytic extraction load.

The model rejects the simple stimulus-transfer equation:

MeaningResponse = f(stimulus)

and replaces it with:

MeaningResponse = f(stimulus_geometry, semantic_access, decoder_availability,
 emotional_integration, extraction_load, confound_burden,
 prior_state_space)

48.2 Branch-level components

For branch *b*, define semantic access:

SA_{*b*} = 1 - mean(AudioComprehensionDifficulty_{*b*},
 SpeakerVolumeDifficulty_{*b*},
 ExternalNoiseIntrusion_{*b*},
 AudioVideoSyncProblem_{*b*}) / 9

Decoder availability:

$$DA_b = \text{mean}(\text{Meaning_b}, \\ \text{Absorption_b}, \\ \text{EmotionalAfterglow_b}, \\ \text{StoryActiveWashout_b}) / 9$$

Emotional/integrative impact:

$$EI_b = \text{mean}(\text{Absorption_b}, \\ \text{EmotionalAfterglow_b}, \\ \text{StoryActiveWashout_b}) / 9$$

Extraction load:

$$X_b = \text{max}(\text{TaskExtractionLoad_core_b}, \\ \text{OverrideTaskBurden_b}) / 9$$

Confound burden:

$$C_b = \text{max}(\text{ConfoundBurden_core_b}, \\ \text{DetailedConfoundMean_b}) / 9$$

48.3 Gate equation

The branch-level decoder-gate estimate is:

$$D_gate_b = \text{sigmoid}(2.0*SA_b + 1.4*DA_b + 1.1*EI_b \\ - 1.2*X_b - 1.8*C_b - 0.75)$$

where:

$$\text{sigmoid}(x) = 1 / (1 + \exp(-x))$$

High D_gate means the branch had good semantic access, strong subjective availability/impact, and tolerable task/confound burden. Low D_gate means one or more gates were closed.

48.4 Gate Dissociation Index

The Gate Dissociation Index compares Target and Override:

$$GDI = \text{mean}(SA_Override - SA_Target, \\ EI_Target - EI_Override, \\ X_Override - X_Target, \\ C_Target - C_Override)$$

A high positive GDI means:

Override had better semantic access,
Target had stronger emotional/integrative impact,

Override had greater extraction load,
Target had greater confound burden.

This is the signature of a gate-dissociation run.

48.5 DGA labels

TARGET_GATE_OPEN_PEAK_PROFILE:

Target gate is high, Target specificity is high, and MRED-Peak is supported.

TARGET_GATE_OPEN_RESOLUTION_PROFILE:

Target gate is high, resolution/afterglow/Baseline2 evidence is strong.

DECODER_GATE DISSOCIATION_AUDIO_ACCESS_SPLIT:

semantic access and emotional impact split across Target and Override.

FEATURE_PROXY_RECOGNITION_DOMINANT_GATE_UNRESOLVED:

older run lacks PRAYCG2.0 reports/Baseline2; only feature-level interpretation is possible.

NO_CLEAR_GATE_OPENING:

neither Target nor Override has sufficient availability or specificity.

48.6 DGA is secondary

DGA does not replace A-MRED. DGA explains why A-MRED may pass, fail, or split. It belongs in the secondary tier:

PRIMARY:

Timing/QC -> CET-R -> A-MRED / MRED-Peak-Resolution

SECONDARY:

NIP / TTI / NUPI / DGA

EXPLORATORY:

KHT-topo / NAST / EET / MRED-ITP / OCM / RSM / CVB / Squint

49. v1.1 update: state-space decoder formulation

49.1 Persistent learned state-space

The decoder-gate update implies that a person does not enter the chamber as a blank slate. The viewer arrives with a learned state-space:

$P_state(t)$ = [autobiographical traces, semantic priors, attachment models,
current-life relevance, prior exposure, affective schemas]

Incoming stimulus geometry is encoded as:

$\phi(u(t))$

Meaning recognition is then:

$MR(t) = D_gate(t) * \text{sim}(\phi(u(t)), P_state(t))$

Integration requires later availability:

$ENC(t + \tau) = MR(t) * A_reg(t + \tau) * [1 - X_extract(t + \tau)] * Q_carry(t + \tau)$

49.2 Update equation

If an event is integrated, the state-space may update:

$P_state(t+1) = P_state(t) + \eta_update * MR(t) * ENC(t+\tau) * Relevance(t) - \text{decay/noise}$

This is a human-scale topology equation, not a cellular proof.

49.3 Consequence for stimulus interpretation

A low-bandwidth cue can have large physiological consequence if it fits a high-relevance internal attractor:

LargeResponse possible when $\text{sim}(\phi(u), P_state)$ is high even if sensory bandwidth is low

This explains why partial semantic access may still open a strong decoder gate in a familiar or personally resonant stimulus.

50. v1.1 update: minimal confirmation-grade decision rule

A run should be called confirmation-grade only if all of the following are true:

FrameVerifiedLockedAnchorFileLoaded = TRUE
AnchorHashRecordedByRunner = TRUE
ALS_Pulse_PASS_Control_Target_Override = TRUE
TargetOverrideHashMatch = TRUE
ControlGeneratedFromCueEmbeddedTarget = TRUE
ControlAudioQC = PASS
StimulusFingerprintAvailable = TRUE
CET_R_DoesNotExplainEndpoint = TRUE
ArtifactGate = PASS

ConfoundGate = PASS

A_MRED_or_MREDPeakResolution = PASS

SelfReportNotContradictory = TRUE

If any key component fails, label the run using the correct caution:

estimated-anchor pilot

operator-frame-verified but ALS-cautioned

runner-registered but audio-confounded

conceptually predeclared / not runner-registered

PRAYCG-lite / no sensory-control branch

The point is not to make confirmation impossible. The point is to make confirmation auditable.